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RESEARCH-ARTICLE

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# Improving Critical Thinking in Software Development via Interdisciplinary Projects at a Most Diverse College

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## ABSTRACT

The software industry has been seeing a steady growth worldwide. However, the quality of the developed software is tightly related to the supply of skilled and capable software developers who are able to cope with many challenges and maintain a high level of critical thinking during the software development life cycle. Software skills are usually gained in upper level software development courses in undergraduate IT majors. We have been applying an active learning methodology for teaching critical thinking in the classroom. Our college is one of the most diverse colleges in the southeast region, which makes our classroom a good model of national diversity. In assessing critical thinking, the traditional and generic approach is to measure skills that are universal and subject-agnostic. In this study, we report that the universal skill assessment for software development is ineffective in measuring students' growth based on inconclusive testing results and a weak test-retest reliability score. We suspect that students were unmotivated by several factors, which includes students being passive listeners and the subject being unrelated to software. In addition, we report a significant potential for developing domain-specific critical thinking exercises and testing that could serve to train the students and to assess their skills at the same time. We would like to emphasize that our conclusion is independent of software development, and could be generalized to develop exercises for other subjects.

## CCS CONCEPTS

• **Social and professional topics** → **Software engineering education; Student assessment; Information technology education;**  
• **Applied computing** → **Education; Software and its engineering** → **Designing software; Programming teams; Agile software development; Documentation.**

## KEYWORDS

software development; software engineering; teaching; critical thinking; CAT; project based learning; PBL

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## 1 INTRODUCTION

Educators agree that critical thinking (CT) is an essential skill, while its effective education at the college level has recently been a topic of intense debate [1, 2, 15, 17]. In the information technology (IT) field, two major academic organizations, the Association for Computer Machinery (ACM) and Institute of Electrical and Electronics Engineers (IEEE), suggested that undergraduate IT programs should provide 5% science education to improve critical thinking [27]. As a subfield of IT, software development and engineering (SD) is no different, and requires learning to solve complex critical thinking challenges for students and practitioners alike [6, 10, 23].

The field of software technologies has been seeing a steady growth worldwide recently. The 2020 LinkedIn Emerging Jobs Report [21] lists "Full Stack Engineer" as the #4 top emerging job in the U.S. list with a 35% annual growth. These job roles may require candidates to know several different programming languages, frameworks, and cloud technologies [14]. And each of these technologies constantly change, become obsolete, and conflict with each other, requiring newcomers into the field to have, not only a simple understanding of programming, but critical thinking skills to make a deliberate selection among possible solutions for the problem in hand based on deep knowledge in the discipline [6].

Critical thinking is a broad concept that includes the following: judging the credibility of a source, identifying needs for clarification in determining aims for a project, evaluating evidence that supports claims, and applying strategies and tactics [9]. There are many current trends in software development whose outcomes can be improved with more CT, such as the Agile philosophy that is changing how software is produced in the industry [23]. But also other, everyday aspects of SD can benefit from more CT, such as those that map into the definition above: understanding customer intentions, solving complex problems that require non-linear thinking, and providing optimal solutions despite conflicting requirements [6]. Engaging SD students into applying more CT can be achieved by emulating real-life complexity in class, for instance by taking a project-based learning (PBL) approach [11], more specifically by employing a *software engineering practicum* [18]. But the

question remains to how should one measure change in CT skills after such educational interventions? Existing CT assessment instruments are focused on measuring a *universal* type of CT skill, although there are who oppose to this idea and believe that CT can be domain-specific skill [17, 19].

### 1.1 Universal versus domain-specific critical thinking

In defining critical thinking, there is a disagreement over whether it is more of a universal or a domain-specific ability (see [17, 19] for a summary). Some scholars define CT as in the definition in the previous section, as a universal ability that can be applied across domains and problems, while others suggest that it requires knowledge and tools specific to a problem domain. Supporters of the latter claim that this domain knowledge is neither transferable nor universal. One argument is that the type of critical thinking required for specific tasks (e.g., playing basketball, finding a way home) may not benefit from a general CT ability (e.g., analyzing arguments). A second argument is that CT that applies to specific fields such as medicine may not translate to other, general domains. Despite these distinctions, the two sides of the argument agree that both universal and domain-specific CT have their own merits. Even then, would using a universal CT assessment instrument be the right choice for detecting an improvement when we are teaching to improve CT specific to the SD domain?

In this study, we used a universal CT assessment tool, the CAT (Critical Thinking Assessment Test), a standard instrument developed at the Tennessee Tech University [24], to measure critical thinking skill gains in a SD course that had a PBL component supervised by external clients. Here, we report our methodology and results, and make a recommendation for a CT teaching and assessment method specific to the domain of SD, which may be more productive. Other studies also arrived at a similar conclusion by offering software development-specific exercises for improving critical thinking skills [6]. However, in this paper, we report detailed evidence showing that a universal assessment tool may be ineffective.

## 2 METHODOLOGY

### 2.1 A highly diverse student population

This study was performed at Georgia Gwinnett College (GGC), which is an *open-access* undergraduate institution that is known for serving an ethnically and racially diverse student population. It has been ranked #1 in ethnic and racial diversity in the Southern Regional Colleges category by the U.S. News & World Report for the fifth consecutive year as of Fall 2018 [7, 28]. Mundie and colleagues [22] concluded that GGC is "an excellent laboratory to investigate gender/ethnicity questions" in a study where they investigated education concepts, such as course repeats, mastery, equity, and institutional STEM culture. Therefore, the results of our study likely present a good case study for highlighting performance of ethnic and racial minorities in the U.S. [4, 22]. Table 1 shows that our SD student population is similarly diverse as the average U.S. STEM field students. Gender of participants was consistent with U.S. average of IT student body [20]. This diversity combined with GGC's fast growth in a short time presents special challenges

|                           | US    | STEM | GGC  | SD    |
|---------------------------|-------|------|------|-------|
| Female                    | 50.8% | 45%  | 58%  | 18.8% |
| Hispanic/LatinX           | 18.3% | 11%  | 21%  | 21.2% |
| White                     | 72.2% | 60%  | 30%  | 61.4% |
| Black/African American    | 12.7% | 26%  | 32%  | 18.9% |
| Asian                     | 5.6%  | 13%  | 10%  | 11.3% |
| Am. Indian/Alaskan Native | 0.9%  | 0%   | 0.1% | 0%    |
| Multirace/unknown         | 3.4%  | 8%   | 7%   | 17.6% |

**Table 1: Gender and diversity compared between populations of U.S. Census Bureau [4], U.S. STEM degrees [22], GGC college-wide [22], and in the SD classes ( $n = 48$  students answered on CAT pre-tests from multiple sections of the SD2 course). Gender (female student percentage) and ethnicity (hispanic/latinX) categories were measured outside the single-race percentages.**

in teaching and evaluating students' critical thinking in general and in our case, for software development. Our student body is non-traditional because there are many who are first generation college students and others who are employed outside of the college as either part-time or full-time, and preoccupied with multiple family responsibilities. Therefore, the results presented here are most likely impacted by our students' population.

### 2.2 External client-led, app building projects

Our approach employs project-based learning (PBL) in class. In PBL, students actively explore and try to surmount real-world challenges and problems using, as a basis, past and recently acquired knowledge. Students deeply explore the problem, learn by practice and experiment, and hone their critical thinking skills. PBL is a student-centered active learning/inquiry based learning pedagogy. A meta-analysis [26] found that when implemented well, PBL can increase long-term retention of material and replicable skills, as well as improve teachers' and students' attitudes towards learning.

For the software development and engineering fields, real-world projects and case studies increase effectiveness of education [11]. Practical knowledge (which PBL engenders) is crucial preparation for students in the real-world career path, which requires a very steep experience for current tools and computer languages.

In Fall 2016, we started offering an alternative to the traditional faculty-led software development course by piloting a PBL approach in our Software Development 2 (SD 2) course. The traditional approach involved the entire class working on the same project. Our approach was more expansive by developing novel products from scratch. Ideas of products for further elaboration by the students were generated by faculty outside the course, who served as clients and mentors throughout the project. Students worked in small groups to convert a client-proposed idea into an application usable by the campus community. To date, the pilot has run for 8 semesters yielding several useful software projects (see <https://soft-eng-practicum.github.io/> for a list of projects and <https://youtu.be/viJMcoy2kl0> for a documentary film).

All pilot software products resulted from in-depth student research. The most notable project is a campus map web application that allows students and faculty to find rooms on campus quickly

and efficiently (<http://ggcmaps.com/>). The project resulted from an interdisciplinary collaboration between SD 2 students and an Arts professor in Fall 2016 (author CAM). Since then, several teams have progressively improved this product, making it into a user-friendly interactive site that is a substantial improvement over the existing PDF-based static maps on the school's website. Students used the Javascript language as a single-page web app (not taught in any course previously taken), manually drew building and campus locations, and created a custom web development framework and novel method to select and highlight rooms on an online display. Throughout the duration of this project, the students have also developed a parking map, a building safety map, and have continued to update the map as our campus expands.

Other noteworthy outcomes resulted from collaborations with the chemistry discipline having author TG as client. Students developed several chemistry learning game applications with the aim of engaging students differently with chemistry related nomenclature and reaction knowledge via a fun and attractive game play and score rewards. These chemistry nomenclature games are ready to be piloted in gateway chemistry courses. Furthermore, several other software projects were presented at local events for physics and chemistry disciplines, and more recently with outside not-for-profit organizations.

By using this project-based instruction method, we aimed for students to improve class engagement and learning satisfaction, to gain experience in working on real-world applications, and to improve student success rate in getting a job in the SD field. Having successfully shown that students can work independently on open-ended projects, we continued to provide students novel, client-developed challenges requiring a solution in the form of an application. Students chose from a selection of client ideas which presumably provide a level of ownership that leads to increased engagement toward completing the project. Since students require to use and improve critical thinking skills as a corollary to the ownership and investment, we started to assess whether students' critical thinking skills showed improvement as a result of the implementation of this PBL exercise.

### 2.3 Assessing universal critical thinking skills

For assessing universal critical thinking skills, several instruments exist, such as the Critical Thinking Assessment Test (CAT) [24], the Assessment of Critical Thinking Ability Survey (ACTA) [29], and the Halpern Critical Thinking Assessment (HCTA) [5, 15]. There are also assessment instruments that furthermore address problem solving in upper level computer science majors [8]. In the present study, we have selected CAT because it is a validated universal CT assessment instrument with no known gender bias or cultural unfairness [24]. In addition, it has been used nationally for more than ten years on more than 30,000 students [16]. We aimed to measure CT skill change during a semester, by administering the CAT instrument twice: pre- and post-treatment, one at the beginning of the semester, and one at the end, as suggested in previous studies [16]. The tests were administered by GGC's NSF-funded Course-embedded Undergraduate Research Experience (CURE) grant [3] administration office staff in absence of this study's investigators. The tests were collected, shuffled together with tests from other

|           | course    | # sections | # tests | # pre+post |
|-----------|-----------|------------|---------|------------|
| control   | ITEC 3860 | 2          | 41      | 14         |
| treatment | ITEC 3870 | 4          | 73      | 23         |

**Table 2: For the control and treatment groups assessed with the CAT test, the table indicates the course where assessment was performed (ITEC 3860 Software Development I vs ITEC 3870 Software Development II), number of course sections assessed (# sections), number of properly scored tests (# tests), and number of students who were present both in pre- and post-testing sessions (# pre+post).**

studies, and then scored by volunteer faculty referees in group sessions. Each CAT test question was independently scored by at least two separate referees, and a third if there was a disagreement. After scoring each question, tests were rotated, such that a single student's test was scored by multiple referees to prevent biased scoring. Referees and session leaders at our institution are regularly evaluated by an external institution for reliability of their scores, and have been deemed to be satisfactory for the past several years. Investigators were instructed to neither share scores, nor discuss, the test with the students to prevent affecting the post-test result. However, we consider this instruction as one of the weaknesses of CAT as reported anecdotally by students (see Discussion). This method of assessment was reviewed and approved by the GGC Institutional Review Board committee (GGC IRB #17082).

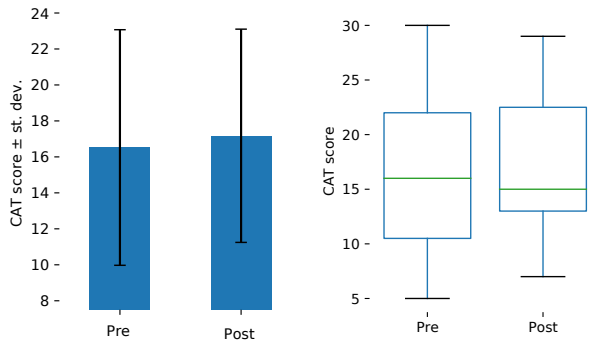
To test the effectiveness of our PBL approach in increasing CT skills, we performed two types of comparisons. First, we employed pre- and post-testing to show the change during treatment periods of one semester. To test the validity of these results, we also compared the treatment class sections to control sections that do not employ the PBL approach, and instead use a traditional lecture format. Our hypothesis was to expect a larger per-semester increase of CT skill in the treatment group. The number of tests scored for different conditions are summarized in Table 2. The treatment group consisted of two course sections for each of the Fall 2018 and Spring 2019 semesters (total of four sections). The control group consisted of two sections of the precursor SD course in Fall 2019.

Outside of our pre- and post-testing scheme, our institution also collects CAT results across its student body. We have compared our results to baseline data from this dataset of 1,791 CAT tests. These tests were collected from students at different years and majors (not just IT). Statistical analysis and breakdown of these test results are presented next.

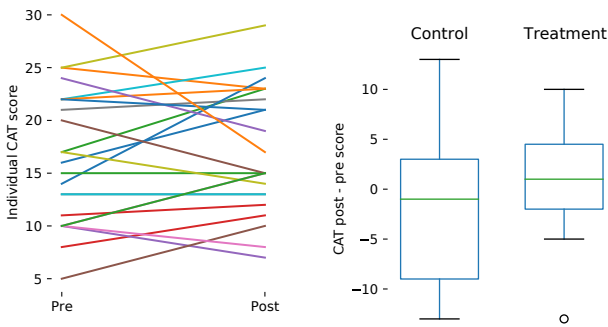
## 3 RESULTS

### 3.1 Paired pre- and post-test CAT scores

We had previously assessed [13] critical thinking abilities with data from two semesters of the SD 2 course, and found a 25% increase in average CAT scores, which was statistically significant ( $n = 49$ , 2-sample two-tailed t-test with unequal variance,  $p < 0.05$ ). In the more rigorous case of only considering paired results with students who took both the pre- and post-tests, we still observed an 11% increase.



(a) Distributions of pre- versus post-test scores from a total of  $n = 23$  students from the treatment group compared as mean  $\pm$  standard deviation (left) and as boxplots (right). Boxplots were used to describe statistics of the different student test score distributions. In these, each box extends from the 25 to the 75 percentile value of the data, with the median marked inside with a green line. The whisker lines extend from the edges of the box to 1.5 times the inter-quantile distance (size of the box).

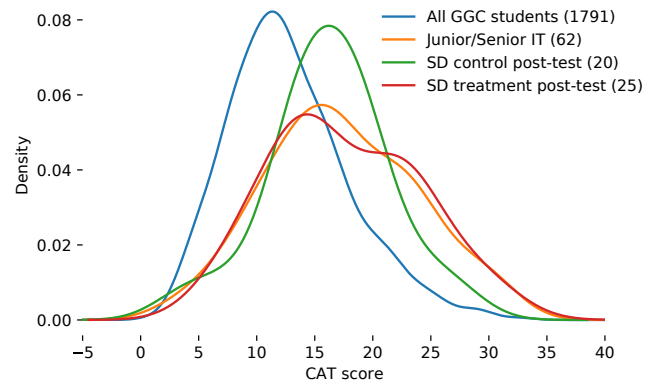


(b) Line plot showing individual student performance between the pre- and post-treatment scores ( $n = 23$ ).

(c) Boxplot distributions of control ( $n = 14$ ) versus treatment ( $n = 23$ ) of pre-test score subtracted from post-test.

**Figure 1: Paired CAT pre- and post-test scores show a positive treatment effect.**

In the present study, we added data from two more treatment sections, which increased the total number of paired CAT test students from 16 to 23. The paired scores still increased on average, albeit less than before, by 3.9% (Fig. 1a, left). We expected the added data to increase the statistical power of our previous conclusions, but instead the results became inconclusive, with the median of pooled CAT post-test score becoming lower than that of pre-test (Fig. 1a, right). The reduced improvement could be explained by the lower post-test scores of the several newly added students – indicating they had lost critical thinking ability during the semester, and some of them in large amounts (Fig. 1b). One outlying data point representing a student dropped from, the best score in this group, of 30, down to 15, which is below our average. Our reduced success motivated us to compare these results to control data.



**Figure 2: Normalized distribution density plots compared between groups of students who took the CAT test showed a progression of increased scores for our target groups. The number of participants in each group is shown in parentheses.**

The control group from the pre-cursor software development course had a negative pre- to post-test difference (on average students lose critical thinking abilities over the course of the semester), while the treatment group median was positive, which was encouraging (Fig. 1c). When comparing averages instead of median, the treatment effect was an increase of 9.72%. As expected from the increases in the moments, the whole distribution also moved to a more positive range, with the student who dropped from 30 down to 15 appearing as outlier outside of the boxplot and whisker range. However, the increase from the control to treatment groups was not statistically significant (neither using a parametric Student's  $t$ -test or a non-parametric Kolmogorov-Smirnov test). But how does this skill increase in treatment effect compare to the expected change in critical thinking abilities during college?

The CAT test have been employed for measuring critical thinking of all majors at our institution. Therefore, we compared the CAT scores of test takers from different groups to confirm that our score distributions were within reasonable ranges (Fig. 2). Density plots allowed us to compare the peaks of the distributions regardless of the changing number of participants. Out of the 1,791 students from all majors and years at GGC, we saw a positive shift in scores for the junior and senior IT major students, who are closest to our subject groups. The IT students group had a wider score distribution, which was similar in shape to our SD students treatment group. In contrast, the control group was shaped narrowly, similar to the general student body, but shifted positive, presumably because of the seniority. Finally, we see a secondary bump on the positive side of the treatment group, which must have contributed to the positive treatment effect calculated above.

Our % CAT score increases could be transformed to a test-independent scale, by comparing it to the standard deviations of CAT scores in our institution's general student body, which was 5.23. Therefore, the change from pre- to post-test in our treatment group ( $n = 23$ ) corresponded to an increase of 0.12 standard deviations. And change between the post-tests of the prior control course ( $n = 20$ ) and the

treatment course ( $n = 25$ ) corresponded to a larger increase of 0.27 standard deviations.

We attribute the apparent "loss of critical thinking abilities" between pre- and post-tests and the lack of statistical power in our nevertheless positive results to the generality of the CAT instrument based on anecdotal feedback we received from students. Students preferred not to take the test despite the extra credit offered for being present during testing day, resulting in low participation numbers, and performed worse on the post-test when they realized it is the same test as before. Some students told instructors that they found the test not a good use of their time because it was not related to software development. Therefore, we proposed a critical thinking exercise and assessment that is constructed with software development concepts and applies to the domain.

### 3.2 Proposed software development-specific critical thinking exercise and assessment

Critical thinking is an integral part of many tasks in the software development life cycle. For example, to effectively communicate with stakeholders, understand requirements, propose effective solutions and to judge and choose deliberately from multiple solutions according to the surrounding circumstances. We suggest that designing a critical thinking exercise involving these concepts that can help us measure students' skill levels could be more effective and motivating than universal CT tests. To this end, we designed an exercise (Table 3) that targets software development in web and mobile platforms and provides both basic knowledge and constraints to work with, and can be scored via two questions that do not have straightforward answers. First one requires students to evaluate the knowledge and the constraints to propose several viable solutions. The second question requests students to evaluate the given constraints and propose changes that could affect the previously proposed solutions. Our pilot application of this exercise in class was encouraging as the students were intrigued and many of them devoted significant time to solve the questions. In Fall 2018 and Spring 2019 semesters, 34 students from the same treatment sections for which we presented CAT results, participated in different versions of this exercise, although they were not scored in a consistent fashion to draw strong conclusions. In particular, the exercise forced students outside of their comfort zone because the constraints prevented them from suggesting solutions with which they were familiar. Unlike the CAT test, they were able to address their frustration in not answering some of the questions by discussing possible solutions with their peers even many days later since it is based on the course subject. Our observation was that the exercise sparks their interest and becomes an eye opener for new technologies and solutions. Another positive outcome was expanding students' expertise during the course. Our PBL approach makes students work in teams for their own projects, they specialize on one type of technology during that semester. However, this exercise exposes students to search for a general software technology solution, which is unexpectedly difficult to them. We are including this study as an example to build upon future SD-specific CT assessments, which could also address software development lifecycle and process subjects.

Answer the questions below based on the knowledge and requirements provided.

*Knowledge:*

- Application server is a monthly-subscription where you can run a back-end web service
- Database server is a monthly-subscription online service that allows accessing a database with an IP address
- Github Pages is a free web publication service that can serve static content
- MongoDB is a noSQL database software that serves JSON documents
- MongoDB Atlas is an online service that offers a MongoDB instance for free up to 500MB
- Javascript can be used to access Mlab and MongoDB
- MySQL is a free, open source DBMS software helping to host a database server on an application/database server
- Altvista is an online service to host web applications and MySQL databases, but it is slow for free usage
- Mobile applications can be downloaded from online markets for free, only with a one-time, small submission fee
- Mobile apps larger than 100MB are not easily installed by users
- Web pages can be viewed on mobile device browsers
- Client-side Javascript code can be hosted without the need for an application server

*Requirements:*

- Client wants app to give info to their user base, which is large (100s)
- Client has more data than 500MB
- Most data doesn't change, but small parts change daily
- Client is non-profit and wants mobile access
- Can't afford application or database server
- Mission critical information needs to be read by clients fast

*Question(s):*

- (1) List up to three potential technological solutions for this client that satisfy ALL of their requirements
- (2) What three changes in requirements will have an impact on the solution provided? Be specific on the change and the new solution for each.

**Table 3: An example critical thinking exercise to practice and measure skill levels specific to software engineering.**

## 4 DISCUSSION

### 4.1 Measured improvements in critical thinking abilities

Overall, we observed a 3.9% increase of CAT scores between pre- and post-test in our treatment sections. Huber and Kuncel [17] compared different studies that report gains in critical thinking by using various tests, by normalizing them as standard deviations. The gains from different studies were in the range of 0.12–0.18 standard deviations (also see [1]), except in a case of a philosophy

course study that reported 0.78 standard deviations gain per semester. Therefore, the increase of 0.12 that we report from pre- and post-treatment seem to be on par with the average gain of critical thinking per a regular college semester, without considering any effect from our treatment. The increase from the control group was larger, at 0.27, suggesting a gain from our treatment.

#### 4.2 Why testing for universal critical thinking was ineffective

Based on our interactions with the participating students and feedback we received, students were clearly not motivated to take the exact same test twice, especially when the content was completely unrelated not only to the course content, but also to the major and specialization of our information technology students. Some students were troubled that they never received feedback about their performance on the pre-test. These explanations are consistent with the post-test student scores that were lower than their pre-test scores, suggesting that they abandoned the test midway or did not put similar effort in answering the questions. Therefore, if we cannot interpret the test results as directly explaining critical thinking skills, but they are confounded by these issues, then the effectiveness of this type of testing needs to be questioned.

Another problem was the popularity of the CAT instrument, which was an asset that allowed us to easily acquire and administer it at our institution. However, it also affected our results. We heard anecdotal evidence of some students having taken this test previously at our institution or, even before coming to college, in high school. One student even claimed taking it for the fourth time. Since the CAT is always the same, with the exact same questions, the effectiveness of the later testing sessions becomes questionable. Stein et al. [25] investigated the effect of repeated testing by applying CAT in a control setting with a social sciences course of 16 students. They applied the pre- and post-test scenario with a 13-week gap in between, and as a result, they calculated a *test-retest reliability coefficient* of 0.6 with a certainty of  $p < 0.1$ . We would expect a high reliability in this control scenario, but the coefficient value of 0.6 corresponds to "questionable reliability" [12]. When we calculated the same score for our control group, we found a negative correlation because of our inconsistent test results. Therefore, two of more repeats of this test may yield unreliable results.

The CAT results from both the general junior/senior IT student body, and post-test results from our treatment sections showed a non-unimodal distribution with a secondary bump. It would be interesting to investigate the reason for this bump, and whether it relates to the perceived ineffectiveness of the test.

From an investigator's and instructor's perspective, the CAT test was also unsustainable. Not only it is a proprietary test that requires purchasing individual copies for each student, but it also took more than 60 minutes to administer the test. With external proctoring and instructions, it was closer to 90 minutes. Therefore, administering this twice a semester with a pre- and post-testing protocol resulted in a significant loss of class time that could have been spent on the class topic. This was more severe especially for our non-traditional students who are already pressed for time. Since the test content was not relevant, it was not even practice, which

discourages instructors from using it in class when they should be preparing students for a job in the industry.

There are other potential universal critical thinking tests such as the Assessment of Critical Thinking Ability Survey (ACTA) [29], and the Halpern Critical Thinking Assessment (HCTA) [15], which may offer improvements on some of these issues, but they will still not match the course topic. In particular, HCTA has been validated to predict real-world outcomes in multiple domains, such as education, health, law, finance, and interpersonal relationships [5].

#### 4.3 Our software-specific critical thinking exercise was received well by students

The exercise that we developed in this paper aims to be specific to the software development and engineering fields, pushing students to exercise their critical thinking and problem solving skills. We identify the ingredients of the critical thinking in the field of software development and then produce a critical thinking model that will transform the students to effective participants. The goal from our model is to map these ingredients to teach software development practices and skills in a way that students can judge alternative approaches that they will encounter in their professional career. So far, based on the observation of student answers and feedback, the activity seems to be promising in the classroom. However, for it to be successful in the future, we aim to provide a standard scoring guide and more diverse questions, similar to previously successful approaches [6].

Another promising future direction for the current project is exploring the theoretical foundation to explain the relationship between critical thinking and learning that is rooted in the grounding from higher-order cognitive skills (HOCS) [30]. These theoretical models can explain other essential ways to understand the various learning process and domain-specific processes: team creation, task sharing, task performance, and the educator's role.

As a conclusion, we determined that universal critical thinking instruments take too much time and effort that takes the student away from the topic at hand. Therefore, the students lose interest, which makes the results of these tests inconclusive. Despite the loss of student interest in the test, we reported positive gains from our project-based approach and further developed a more effective CT exercise.

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